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INTRODUCTION

In modern days, Artificial Intelligence (AI) has gained extraordinary heights, allowing many significant contributions to be made in various areas of society, meaning education is no exception. The use of AI in education has changed, and is continuing to change, traditional back-in-the-day teaching techniques and self study/learning, alongside these things, AI helps provide innovative solutions to educational challenges that have been in the long run for quite a while.

ROLE OF AI IN EDUCATION

Among the many contributions AI has made, self study/learning in the field of education is one of the most important. Going back to traditional teaching styles, they are rather ineffective as they present the same thing to everyone, which isn't very diverse. Traditional teachings are

more of a set standard which everyone is expected to meet. With the assistance of AI, individuals such as students can personalize their learning as AI adapts to their strengths and learning styles while also taking in their weaknesses into consideration. The existence of adaptive learning technologies, which use algorithms in AI to take in the individual's educational standing and create lessons that better suit them. Such systems include ones like Carnegie Learning's MATHia, Why2-Atlas, AutoTutor, etc.. Systems like these are known as Intelligent Tutoring Systems (ITS), which basically offer the user a one on one tutoring session as they closely recreate the guidance of an in person tutor. ITS utilizes AI in order to understand the user and quickly give feedback while adjusting the teaching to the student.

AI can also help teachers in school with grading, making the overall process easier, faster, and efficient. This is better known as the automated grading system. Such systems can

grade tests, essays, and assignments ranging in various subjects. Using Natural Language Processing (NLP), these systems can evaluate the given content and provide detailed output. One of the many known and used systems in school is called Turnitin, which not only checks for plagiarism but also looks at the originality of the written content. With such systems in the school, teachers can give more focus into instructional activities rather than read hundreds of essays and evaluate each one for long periods of time.

AI is also enhancing modern day accessibility in education for those who have disabilities. Things such as speech recognition can convert spoken vocal words into written text, (i.e. Google Live Transcribe) allowing those who are hearing impaired a better chance at education. The opposite can also be done as text to speech helps the visually impaired by reading digitally written text out loud. Furthermore, AI can

also break language barriers by translating material in multiple languages.

BENEFITS OF AI IN EDUCATION

The presence of AI has been proven to improve learning outcomes with the providence of personalized learning and quickly detailed feedback. There have been multiple studies that show individuals in AI-based adaptive learning environments outshine those who still learn in a traditional learning environment. Alongside improved performance, AI also automates tasks such as grading, giving teachers to focus more time into the person to person interaction among their students. This creates an increased productive environment at school. AI provides valuable feedback to the student in their performance and their standings in their education. This allows teachers to then use such statistics to

weed out those who need an improved learning system and provide as needed.

AI also provides more accessibility of education to those with disabilities like hearing or vision.

PROBLEMS OF AI IN EDUCATION

While there are many benefits, the use of AI in education can also have some negative outcomes. Using AI in education can create privacy concerns. Analyzing an individual's data by systems can lead to potential breaches of privacy in the later run. It is important to establish strong data protection policies and make sure there is transparency with whichever application is being used. AI technologies require access to things within the digital infrastructure, which might not be available to everyone across the world. There is also a thing called digital divide that can create educational inequalities,

which causes certain individuals in underprivileged areas to be at a disadvantage. Addressing this challenge would mean investment in digital infrastructure and making sure the access to AI-powered educational tools is equitable.

Relying too much in AI within education is also another great problem that may lead to a decrease in human interaction or individuals' thinking skills. Knowing this, it is important to create a balance between using AI and traditional teaching techniques to create a balanced educational experience for individuals in studying environments.

FUTURE OF AI IN EDUCATION

Looking into the coming future, AI has a lot of potential to help in lifelong learning by providing ongoing education opportunities to individuals beyond school. Things like

personalized learning platforms can give education to learners of all ages, helping them gain new skills and knowledge throughout their lives.

AI can also allow increased collaborative learning by connecting people from different parts of the world and enabling real-time collaboration on work such as projects. AI-powered platforms, i.e. Google Classroom and Microsoft Teams, are programmed to stimulate virtual classrooms and group activities, allowing a global learning community.

Ongoing advancements in AI technologies, such as machine learning and natural language processing, might further enhance their influence in education. Future AI systems may even offer more complicated and challenging personalized learning experiences, more intricate grading, and improved education to the disabled with accessibility features.

The usage of AI in education also creates a necessity in training and support for teachers in order to effectively use such systems in schools for the future.

CONCLUSION

Overall, AI is transforming, and will continue to transform, education with various assistance in areas like personalized self learning, automating tasks such as grading and scheduling, and enhancing accessibility to those with disabilities. While there are a few challenges, such as ethical concerns and digital divides, the potential benefits of AI in education are much greater and immense in the long run. By addressing these challenges and embracing the presence and usage of AI, society can create a more inclusive, efficient, and effective educational system. The future of education lies in the balanced use of AI and traditional teaching methods,

ensuring that learners of all backgrounds can thrive in today's century along with many more to come.